

Why $\bar{y}_1 - \bar{y}_0$ is Unbiased for the ATE under Random Assignment

Claim. Under random assignment, $E[\bar{y}_1 - \bar{y}_0] = ATE$.

Definition of unbiased. An estimator $\hat{\theta}$ is **unbiased** for parameter θ if:

$$E[\hat{\theta}] = \theta$$

Here $\hat{\theta} = \bar{y}_1 - \bar{y}_0$ and $\theta = ATE = E[y(1) - y(0)]$.

Step 1: What do we observe in each group?

Treated individuals reveal $y(1)$; control individuals reveal $y(0)$:

$$E[\bar{y}_1] = E[y \mid x = 1] = E[y(1) \mid x = 1]$$

$$E[\bar{y}_0] = E[y \mid x = 0] = E[y(0) \mid x = 0]$$

Step 2: Apply random assignment.

Random assignment means $x \perp (y(0), y(1))$: knowing whether someone was treated tells you nothing about their potential outcomes. Therefore:

$$E[y(1) \mid x = 1] = E[y(1)], \quad E[y(0) \mid x = 0] = E[y(0)]$$

Step 3: Substitute and conclude.

$$\begin{aligned} E[\bar{y}_1 - \bar{y}_0] &= E[y(1) \mid x = 1] - E[y(0) \mid x = 0] && \text{(Step 1)} \\ &= E[y(1)] - E[y(0)] && \text{(Step 2)} \\ &= E[y(1) - y(0)] && \text{(linearity of expectation)} \\ &= ATE && \checkmark \end{aligned}$$

Intuition. The fundamental problem of causal inference is that we never observe both $y(1)$ and $y(0)$ for the same person. But with $x \perp (y(0), y(1))$:

- The treated group is a *random sample* of the population $\Rightarrow$ their average $y(1)$ represents **everyone's** $y(1)$.
- The control group is a *random sample* of the population $\Rightarrow$ their average $y(0)$ represents **everyone's** $y(0)$.

Each group serves as a valid counterfactual for the other, and selection bias cancels exactly.